C-B-002185(Army-GL)
March 31, 1983

INTERIM FEDERAL SPECIFICATION

BREAKFAST BAKE, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by the U.S. Army Natick Research and Development Laboratories, Natick, MA 01760, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers breakfast bake thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
JJJ-S-30	-	Salad Oil, Vegetable
PPP-B-636	-	Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

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(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items
MIL-L-35078	-	Loads, Unit: Preparation of Nonperishable Subsistence Items; General Specification for

Military Standards:

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-900	-	Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Canned Foods for the Armed Forces

(Copies of Military specifications and standards required by contractors in connection with procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture (USDA)

Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59)

(Application for copies should be addressed to the Poultry Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Regulations Governing the Grading of Shell Eggs and United States Standards, Grades, and Weight Classes for Shell Eggs (7 CFR Part 56)

(Application for copies should be addressed to the Poultry Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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U.S. Standards for Grades of Maple Syrup

(Application for copies should be addressed to the Chief, Processed Products Branch, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Grades of Nonfat Dry Milk (Spray Process)

(Application for copies should be addressed to Director, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Condition of Food Containers

(Application for copies should be addressed to Director, Food Quality Assurance Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1, and 6.4).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

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3.2.1 Whole eggs. The whole eggs may be liquid or frozen and shall have been processed in accordance with 7 CFR Part 59, Regulations Governing the Inspection of Eggs and Egg Products. The whole eggs shall have been prepared from egg yolks and egg whites in their natural proportions as broken directly from the shell eggs as evidenced by a USDA Egg Products Inspection Certificate. The whole eggs shall contain no additives, and shall be in satisfactory condition, free of objectionable odors, such as musty, sour, or fruity, and free from foreign material. Liquid egg products shall be held at a temperature of 40°F or lower and shall not be held more than 72 hours from time of pasteurization until the start of formulating the breakfast bake. Frozen whole eggs shall be held at 10°F or lower and used within 240 days. Thawed previously frozen egg products shall be held at a temperature of 40°F or lower and shall not be held for more than 24 hours prior to use in processing.

3.2.2 Bread. Bread shall be commercial, soft crumb, enriched, white bread made by the continuous or conventional method. Other breads shall not be used. The bread shall be free from microbial deterioration such as mold, rope, sourness, musty odor, or any flavor or aroma not typical of soft crumb white bread. The commercial white bread used shall have a moisture content of not greater than 38.0 percent; protein content of not less than 8.0 percent; fat content of not less than 1.4 percent; and carbohydrate content of not less than 50.0 percent.

3.2.3 Salad oil. Salad oil shall conform to type B of JJJ-S-30 and shall be confined to properly refined cottonseed or corn oil or a combination of the two. Oils from other sources shall not be used.

3.2.4 Lecithin. Lecithin shall be of food grade quality. It may be bleached or unbleached and be light yellow to light brown in color. It may be odorless or exhibit a characteristic slight nut-like aroma. It shall be bland in flavor, free from rancidity or any pronounced odor and any foreign objects. It must readily emulsify in an oil and water mixture.

3.2.5 Salt. Salt shall be non-iodized, white, refined sodium chloride with or without anti-caking agents.

3.2.6 Sugar. Sugar shall be refined, sucrose, granulated (fine or extra fine), cane or beet sugar or a combination thereof. The sugar shall comply with the bacteriological requirement of MIL-STD-900.

3.2.7 Milk. Milk shall be nonfat dry milk meeting the requirements for U.S. Extra Grade, processed by low heat, of the U.S. Standard for Grades of Nonfat Dry Milk (Spray Process).

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3.2.8 Maple, syrup, imitation. The imitation maple flavoring and coloring ingredients shall be stable products of such quality that when a syrup is made from not less than 3-7/8 pounds of sugar dissolved in 1 quart of boiling water and flavored with 1/2 ounce of the imitation flavor, the resulting syrup shall have a flavor characteristic of the flavor of maple and shall not have a bitter, flat, weak, or medicinal flavor. The color of the syrup shall be not lighter than light amber nor darker than medium amber as represented by the color standards for maple syrup in the U.S. Standards for Grades of Maple Syrup.

3.3 Breakfast bake preparation. The breakfast bake shall be prepared from the following ingredients:

Ingredient	Percent by weight
Bread	19.36
Water	37.26
Milk, non-fat dry	3.78
Eggs	27.86
Salad oil	4.84
Sugar	6.00
Salt	0.15
Maple syrup, imitation	0.50
Lecithin	0.25
	100.00

NOTE: The moisture content of the bread used in the formula shall be determined before combining with the above ingredients. If the moisture content is determined to be greater than 38 percent, that amount in excess shall be deducted from the percent water in the formulation. If the moisture content is less than 38 percent, adjust the formulation with the required amount of water.

The bread and half of the water shall be mixed to form a thick, smooth mass. The non-fat dry milk shall be rehydrated in the remaining water and gradually blended in with the bread mixture until a smooth, lump free mixture is formed. The remaining ingredients shall be added and mixed until the dry ingredients are dissolved and a uniform mixture is obtained. The temperature of the mixture shall not exceed 40°F at any time prior to filling into the cans. If the mixture is allowed to stand after mixing of the dry ingredients it shall be remixed before filling. In addition, the mixture shall be periodically stirred during can filling to prevent settling.

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3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product such as to conform to the following requirements:

a. The average net weight shall be 108 ounces minimum.

b. No individual can shall contain less than 106 ounces of product.

c. Each tray pack can shall be hermetically sealed under a vacuum equivalent to 10 inches of mercury immediately after filling.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting at 240°F until a sterilization value (F_o) of not less than 5 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material, such as, but not limited to dirt, insect, insect part, hair, wood, paper, paint, glass, or metal.

b. The product shall have a tan surface with a uniform creamy-tan interior.

c. The product shall possess a slightly sweet maple flavor.

d. The product shall have a smooth surface with a moist, dense grainy custard interior.

e. The product shall possess a slight sweet maple odor.

f. The product shall have a dense smooth spongy texture.

g. There shall be no void areas or air pockets measuring greater than 0.5 inch in any dimension.

h. The fat content of the finished product shall not exceed 9 percent.

i. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of the applicable government inspection agency

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3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting it to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this specification. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation and related services shall be borne by the contractor.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1 Ingredient examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, US Grade certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or of any involved product.

4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

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4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 0.1 ounce. If the average net weight is less than 108 ounces or if the net weight of any individual can is less than 106 ounces, the lot shall be rejected.

4.6.2 Product inspection. Eight tray packs shall be selected at random from the lot. The eight tray packs shall be heated at least 15 minutes in 140° to 150°F water, opened, and inspected in accordance with table I.

a. If no defects are found, the lot shall be accepted.

b. If two or more defects are found, the lot shall be rejected.

c. If only one defect is found, a second sample of eight tray packs shall be selected at random from the lot, heated as above, and inspected for the defects listed in table I. If any defect is found in the second sample, the lot shall be rejected.

TABLE I. Product inspection 1/ 2/

Category	Defect
Major	
151	Product surface not tan, with a uniform creamy-tan interior.
152	Product does not possess a slightly sweet maple flavor or odor.
153	Product not uniform in appearance and does not possess a smooth surface with a moist, dense grainy custard interior.
154	Texture of product not dense, smooth and spongy.
155	Presence of void areas or air pockets measuring more than 0.5 inch in any dimension.

1/ The presence of any foreign material (e.g., glass, dirt, insect parts, hair, wood, metal, etc.) foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale, etc.) or foreign color shall be cause for rejection of the lot.

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2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot, (This comparison shall only be performed when deemed necessary by an AMS agent).

4.6.3 Fat content testing. Nine tray packs shall be selected from the lot and distributed as follows:

- Three for laboratory analysis
- Three for submission to the contractor
- Three for retention by the AMS agent as a reserve sample.

The three tray packs for laboratoray analysis shall be individually tested for fat content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, except that preparation of the samples shall be as follows. The three unopened tray packs shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. The lot shall be rejected if the fat content of any individual sample exceeds 9.0 percent. Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratoray analysis indicated that the involved lot will be rejected because of noncompliance with the fat content requirement. When the reserve samples are analyzed, the analysis for fat shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.8 Shipping container examination. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in construction, closure, and reinforcement shall be in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: National stock number, item description, contract number, or date of pack markings missing, incorrect or illegible. Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not specified; (3) When applicable reinforcement with nonmetallic strapping or tape is not used.

4.9 Unit load inspection. Inspection of unit loads shall be in accordance with quality assurance provisions of MIL-L-35078.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 108 ounces of the food product, sealed and thermoprocessed as specified in 3.4 and 3.5. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient weight and temper to protect the product during shipment and storage. The entire inside area of the can shall be coated. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed, closed and reinforced in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pads shall be full length and width less approximately 1/4 inch in each dimension. The box liner and pads shall be fabricated of grade V3c fiberboard.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.2.1 Nonmetallic reinforcement of shipping containers. When specified (see 6.1 and 6.3), the shipping container specified in 5.2.2 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The pads shall be full length and width less approximately 1/4 inch in each dimension and shall be fabricated of class domestic fiberboard.

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5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. For distribution to the Navy and Air Force, when unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT

a. To heat in water: Submerge unopened can in boiling water. Simmer gently 35 minutes. Remove from water and open lid for serving.

b. To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 35 to 40 minutes until contents reach an internal temperature of 165°F.

WARNING: Do not place closed can in oven as heating will increase internal pressure which may cause the can to burst.

YIELD

Approximately 20 servings per tray. Each serving approximately a 2 inch by 3 inch slice.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.3, and 6.4).
(c) Provisions for approved preproduction samples (see 3.6 and 6.4).
(d) Level of packaging and packing required (see 5.1 and 5.2).
(e) When nonmetallic reinforcement is required for level B packing (see 5.2.2.1).
(f) Type and class of unit load when unit loading is required (see 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 Selection of nonmetallic reinforcement. The nonmetallic reinforcement specified in 5.2.2.1 should be selected for shipping containers that are destined for transfer at sea or other specific overseas operations.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55 The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

Preparing activity:

Army - GL

Project No. 8940-A415

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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